

# Lab 01 Propositional Logic

## Objective

Solving exercises from the textbook in chapter 1.1

## Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to:

understand propositional logic concepts and practice the truth tables.

## Lab Requirements

Students allowed using their lecture notes in the lab in order to solve the exercises.

## Lab Assessment

- 1- Divide students to groups and let them to solve the given example.
- 2- Discuss the answers with the groups and write on board the optimal solution.

## Lab Description

1. Which of these sentences are propositions? What are the truth values of those that are propositions?
  - a) Do not pass go.
  - b) What time is it?
  - c)  $2 + 3 = 5$ .
  - d)  $5 + 7 = 10$ .
  - e)  $x + 2 = 11$ .
  - f)  $2^n \geq 100$
2. What is the negation of each of these propositions?
  - a) Jennifer and Teja are friends..
  - b) There is no pollution in New Jersey.
  - c)  $2 + 1 = 3$ .
3. Let  $p$  and  $q$  be the propositions  $p$  : It is below freezing.  $q$  : It is snowing. Write these propositions using  $p$  and  $q$  and logical connectives(including negations).
  - a. It is below freezing and snowing.
  - b. It is below freezing but not snowing.
  - c. It is not below freezing and it is not snowing.
  - d. It is either snowing or below freezing (or both).
  - e. If it is below freezing, it is also snowing.
  - f. Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.
  - g. That it is below freezing is necessary and sufficient for it to be snowing.
4. Let  $p$  and  $q$  be the propositions "Swimming at the New Jersey shore is allowed" and "Sharks have been spotted near the shore," respectively. Express each of these compound propositions as an English sentence.
  - a)  $\neg q$
  - b)  $p \wedge q$
  - c)  $\neg p \vee q$
  - d)  $p \rightarrow \neg q$
  - e)  $\neg q \rightarrow p$
  - f)  $\neg p \rightarrow \neg q$
  - g)  $p \leftrightarrow \neg q$
  - h)  $\neg p \wedge (p \vee \neg q)$ .
5. How many rows appear in a truth table for each of these compound propositions?
  - a.  $(q \rightarrow \neg p) \vee (\neg p \rightarrow \neg q)$
  - b.  $(p \vee \neg t) \wedge (p \vee \neg s)$
  - c.  $(p \rightarrow r) \vee (\neg s \rightarrow \neg t) \vee (\neg u \rightarrow v)$
  - d.  $(p \wedge r \wedge s) \vee (q \wedge t) \vee (r \wedge \neg t)$

6. Construct a truth table for each of these compound propositions.
- a)  $(p \vee \neg q) \rightarrow q$       b)  $(p \vee q) \rightarrow (p \wedge q)$       c)  $(p \vee q) \rightarrow (p \oplus q)$       d)  $(p \vee q) \oplus (p \wedge q)$
- e)  $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$       f)  $(p \oplus q) \rightarrow (p \oplus \neg q)$
7. Suppose that Smartphone A has 256MB RAM and 32GB ROM, and the resolution of its camera is 8 MP; Smartphone B has 288 MB RAM and 64 GB ROM, and the resolution of its camera is 4 MP; and Smartphone C has 128 MB RAM and 32 GB ROM, and the resolution of its camera is 5 MP. Determine the truth value of each of these propositions.
- a) Smartphone B has the most RAM of these three smart phones.
- b) Smartphone C has more ROM or a higher resolution camera than Smartphone B.
- c) Smartphone B has more RAM, more ROM, and a higher resolution camera than Smartphone A.
- d) If Smartphone B has more RAM and more ROM than Smartphone C, then it also has a higher resolution camera.
- e) Smartphone A has more RAM than Smartphone B if and only if Smartphone B has more RAM than Smartphone
8. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.
- a) Coffee or tea comes with dinner.
- b) A password must have at least three digits or be at least eight characters long.
- c) The prerequisite for the course is a course in number theory or a course in cryptography.
- d) You can pay using U.S. dollars or euros
9. Write each of these statements in the form “if  $p$ , then  $q$ ” in English. [Hint: Refer to the list of common ways to express conditional statements.]
- a) It snows whenever the wind blows from the northeast.
- b) The apple trees will bloom if it stays warm for a week.
- c) That the Pistons win the championship implies that they beat the Lakers.
- d) It is necessary to walk 8 miles to get to the top of Long’s Peak.
- e) To get tenure as a professor, it is sufficient to be world famous.
- f) If you drive more than 400 miles, you will need to buy gasoline.
- g) Your guarantee is good only if you bought your CD player less than 90 days ago.
- h) Jan will go swimming unless the water is too cold.